No.	Status	Date
LNDE-RPT-20220906-033-DS-4-RA-102	Issued	2022-09-16

UT Thickness Data Sheet - 4-RA-102

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurement results for line 4-RA-102, located within the ATU-100. The scope of work involves the recording of wall thickness to monitor asset integrity and corrosion rates for rich amine service.

Marcus Chen, acting as the lead UT technician and SNT-TC-1A UT Level II for Larkhall NDE Services Inc., is responsible for the execution of the measurements and the accuracy of the recorded data.

Technical Basis

All thickness measurements were performed in accordance with QSCP-NDT-001 Rev 2, ensuring compliance with site-specific requirements for ultrasonic thickness measurement within the ATU-100, including necessary insulation considerations.

The inspection methodology and equipment calibration adhere to the requirements specified in ASTM E797.

Line Specifications

The subject line, 4-RA-102, is constructed from A106 Gr.B with an outer diameter of 114.3 mm and a nominal wall thickness of 8.56 mm. The minimum allowable thickness (t-min) is 3.81 mm.

Design parameters for this line are established at a maximum allowable pressure of 862 kPa(g) and a maximum allowable temperature of 82 °C.

Inspection Context

This data sheet supports the findings detailed in LNDE-RPT-20220906-033-R0, which covers the 3-year RBI survey for Class 1 sour gas, regenerator overhead, and rich amine services.

The survey focuses on identifying wall loss and calculating remaining life based on the difference between current readings and the 3.81 mm retirement thickness.

Thickness readings, measured 2022-09-16

Line 6-SG-101 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-101-01	7.11	3.56	2019-10-18	5.93	5.82	0.049	0.053	42.6	2025-09-16	
6-SG-101-02	7.11	3.56	2019-10-18	5.51	5.42	0.061	0.052	30.3	2025-09-16	
6-SG-101-03	7.11	3.56	2019-10-18	5.56	5.29	0.067	0.088	19.7	2025-09-16	
6-SG-101-04	7.11	3.56	2019-10-18	5.34	5.19	0.072	0.059	22.8	2025-09-16	
6-SG-101-05	7.11	3.56	2019-10-18	5.04	4.72	0.091	0.100	11.6	2025-09-16	
6-SG-101-06	7.11	3.56	2019-10-18	5.56	5.55	0.057	0.050	34.7	2025-09-16	
6-SG-101-07	7.11	3.56	2019-10-18	6.24	6.16	0.027	0.027	>50	2025-09-16	
6-SG-101-08	7.11	3.56	2019-10-18	5.77	5.57	0.066	0.065	30.6	2025-09-16	

Line 6-SG-102 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-102-01	7.11	3.56	2019-10-18	5.05	4.77	0.089	0.099	12.3	2025-09-16	
6-SG-102-02	7.11	3.56	2019-10-18	5.10	4.64	0.093	0.120	9.0	2025-09-16	
6-SG-102-03	7.11	3.56	2019-10-18	4.92	4.69	0.089	0.080	12.8	2025-09-16	
6-SG-102-04	7.11	3.56	2019-10-18	4.90	4.62	0.089	0.085	11.9	2025-09-16	
6-SG-102-05	7.11	3.56	2019-10-18	6.07	5.96	0.044	0.035	>50	2025-09-16	
6-SG-102-06	7.11	3.56	2019-10-18	6.03	5.79	0.054	0.041	41.6	2025-09-16	

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	2019-10-18	7.29	6.83	0.064	0.152	19.9	2025-09-16	
4-RA-101-02	8.56	3.81	2019-10-18	7.39	6.79	0.066	0.174	17.1	2025-09-16	
4-RA-101-03	8.56	3.81	2019-10-18	6.64	5.98	0.099	0.253	8.6	2025-09-16	
4-RA-101-04	8.56	3.81	2019-10-18	7.54	7.21	0.054	0.120	28.4	2025-09-16	
4-RA-101-05	8.56	3.81	2019-10-18	7.18	6.73	0.071	0.165	17.7	2025-09-16	
4-RA-101-06	8.56	3.81	2019-10-18	6.94	—	—	—	—	—	inaccessible – no reading
4-RA-101-07	8.56	3.81	2019-10-18	7.68	7.50	0.041	0.097	38.0	2025-09-16	
4-RA-101-08	8.56	3.81	2019-10-18	6.45	5.55	0.118	0.267	6.5	2025-09-16	
4-RA-101-09	8.56	3.81	2019-10-18	7.68	7.37	0.045	0.115	30.9	2025-09-16	
4-RA-101-10	8.56	3.81	2019-10-18	7.55	7.19	0.056	0.126	26.8	2025-09-16	
4-RA-101-11	8.56	3.81	2019-10-18	6.66	6.07	0.098	0.232	9.7	2025-09-16	
4-RA-101-12	8.56	3.81	2019-10-18	7.03	6.41	0.213	0.213	12.2	2025-09-16	

Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-01	8.56	3.81	2019-10-18	7.58	7.42	0.042	0.091	39.7	2025-09-16	
4-RA-102-02	8.56	3.81	2019-10-18	6.96	6.43	0.081	0.184	14.3	2025-09-16	
4-RA-102-03	8.56	3.81	2019-10-18	7.05	6.56	0.075	0.150	18.3	2025-09-16	
4-RA-102-04	8.56	3.81	2019-10-18	7.04	6.64	0.072	0.164	17.3	2025-09-16	
4-RA-102-05	8.56	3.81	2019-10-18	7.06	6.56	0.075	0.168	16.3	2025-09-16	
4-RA-102-06	8.56	3.81	2019-10-18	7.15	6.65	0.070	0.164	17.3	2025-09-16	
4-RA-102-07	8.56	3.81	2019-10-18	7.30	6.85	0.065	0.141	21.6	2025-09-16	
4-RA-102-08	8.56	3.81	2019-10-18	6.40	5.86	0.099	0.226	9.1	2025-09-16	
4-RA-102-09	8.56	3.81	2019-10-18	6.78	6.42	0.080	0.170	15.4	2025-09-16	
4-RA-102-10	8.56	3.81	2019-10-18	7.36	6.86	0.172	0.172	17.8	2025-09-16	
4-RA-102-11	8.56	3.81	2019-10-18	7.72	7.33	0.074	0.096	36.8	2025-09-16	

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-01	10.97	4.57	2019-10-18	9.83	9.55	0.055	0.112	44.4	2025-09-16	
6-RA-103-02	10.97	4.57	2019-10-18	8.61	7.97	0.118	0.261	13.0	2025-09-16	
6-RA-103-03	10.97	4.57	2019-10-18	9.49	8.95	0.077	0.168	26.0	2025-09-16	
6-RA-103-04	10.97	4.57	2019-10-18	8.22	7.30	0.139	0.305	9.0	2025-09-16	
6-RA-103-05	10.97	4.57	2019-10-18	8.33	7.43	0.133	0.311	9.2	2025-09-16	
6-RA-103-06	10.97	4.57	2019-10-18	9.79	9.45	0.059	0.132	37.0	2025-09-16	
6-RA-103-07	10.97	4.57	2019-10-18	9.59	9.15	0.069	0.159	28.8	2025-09-16	
6-RA-103-08	10.97	4.57	2019-10-18	9.73	9.29	0.063	0.158	29.9	2025-09-16	
6-RA-103-09	10.97	4.57	2019-10-18	9.78	9.43	0.060	0.137	35.6	2025-09-16	
6-RA-103-10	10.97	4.57	2019-10-18	8.37	7.68	0.126	0.275	11.3	2025-09-16	
6-RA-103-11	10.97	4.57	2019-10-18	8.91	—	—	—	—	—	inaccessible – no reading
6-RA-103-12	10.97	4.57	2019-10-18	9.52	9.11	0.141	0.141	32.3	2025-09-16	
6-RA-103-13	10.97	4.57	2019-10-18	9.55	9.13	0.114	0.155	29.5	2025-09-16	
6-RA-103-14	10.97	4.57	2019-10-18	8.26	7.34	0.316	0.316	8.8	2025-09-16	

Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-01	7.62	3.56	2019-10-18	6.05	5.92	0.063	0.064	37.0	2025-09-16	
3-RV-101-02	7.62	3.56	2019-10-18	5.97	5.81	0.066	0.076	29.7	2025-09-16	
3-RV-101-03	7.62	3.56	2019-10-18	5.24	4.98	0.096	0.099	14.4	2025-09-16	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-04	7.62	3.56	2019-10-18	5.58	5.39	0.083	0.074	22.1	2025-09-16	
3-RV-101-05	7.62	3.56	2019-10-18	6.11	5.91	0.066	0.068	34.4	2025-09-16	
3-RV-101-06	7.62	3.56	2019-10-18	5.55	5.39	0.083	0.086	21.2	2025-09-16	
3-RV-101-07	7.62	3.56	2019-10-18	5.07	4.80	0.117	0.112	10.6	2025-09-16	
3-RV-101-08	7.62	3.56	2019-10-18	5.86	5.81	0.075	0.076	29.7	2025-09-16	

Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-01	5.54	2.79	2019-10-18	4.16	3.91	0.061	0.067	16.8	2025-09-16	
2-RV-102-02	5.54	2.79	2019-10-18	3.86	3.64	0.074	0.056	11.6	2025-09-16	
2-RV-102-03	5.54	2.79	2019-10-18	4.37	4.34	0.043	0.046	34.1	2025-09-16	
2-RV-102-04	5.54	2.79	2019-10-18	4.59	4.47	0.042	0.039	40.1	2025-09-16	
2-RV-102-05	5.54	2.79	2019-10-18	4.78	4.59	0.037	0.058	31.2	2025-09-16	
2-RV-102-06	5.54	2.79	2019-10-18	4.27	4.05	0.064	0.033	19.6	2025-09-16	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-01	8.56	3.81	2019-10-18	7.15	6.61	0.072	0.167	16.8	2025-09-16	
4-RA-104-02	8.56	3.81	2019-10-18	7.31	7.04	0.057	0.123	26.3	2025-09-16	
4-RA-104-03	8.56	3.81	2019-10-18	7.94	7.70	0.074	0.088	44.2	2025-09-16	
4-RA-104-04	8.56	3.81	2019-10-18	7.64	7.31	0.101	0.132	26.5	2025-09-16	
4-RA-104-05	8.56	3.81	2019-10-18	7.57	7.12	0.109	0.158	21.0	2025-09-16	
4-RA-104-06	8.56	3.81	2019-10-18	7.13	6.70	0.071	0.138	20.9	2025-09-16	
4-RA-104-07	8.56	3.81	2019-10-18	6.67	6.04	0.096	0.194	11.5	2025-09-16	
4-RA-104-08	8.56	3.81	2019-10-18	7.46	7.09	0.094	0.109	30.0	2025-09-16	

Sign-off	Name	Function	Date	Signature
Prepared	Marcus Chen	lead UT technician, SNT-TC-1A UT/MT/PT Level II; CAN/CGSB-48.9712 UT Level II	2022-09-16	_____